

Date **1.10.25**

Exam **hw1**

Subject

Importance ☆☆☆☆☆

Review □□□□□



1. (2 points) Let A, B, C be assertions. Use a *truth table* to show that the following assertion is true:

$$[(A \wedge B) \vee C] \Leftrightarrow [(A \vee C) \wedge (B \vee C)].$$

(The other *distributive law* is $[(A \vee B) \wedge C] \Leftrightarrow [(A \wedge C) \vee (B \wedge C)]$.)

A	B	C	$A \wedge B$	$(A \wedge B) \vee C$	$A \vee C$	$B \vee C$	$(A \vee C) \wedge (B \vee C)$
T	T	T	T	T	T	T	T
T	T	F	T	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	F	T	F	F
F	T	T	F	T	T	T	T
F	T	F	F	F	F	T	F
F	F	T	F	T	T	T	T
F	F	F	F	F	F	F	F

With the truth table, all the results validate as equal.
Thus, $[(A \wedge B) \vee C] \Leftrightarrow [(A \vee C) \wedge (B \vee C)]$ is **always True**.

result ... B
condition ... A



2. (2 points) Reformulate each of the following statements, using a complete sentence of the form "If A, then B.":

- (a) No three positive integers a, b, c satisfy the equation $a^n + b^n = c^n$ for any integer n greater than 2.
- (b) Every square complex matrix is the sum of two commuting matrices.

(a) If n is an integer greater than 2, and a, b, c are positive int., then $a^n + b^n \neq c^n$

(b) If M is a square complex matrix, then M can be written as the sum of two commuting matrices.



3. (2 points) Formulate a *contraposition* of each of the following implications:

- (a) If $3n + 11$ is even, then n is odd. **not odd, not even $\rightarrow$ even, odd.**
- (b) If $A \in \mathbb{C}^{n \times n}$ is singular, then there exists a $\delta > 0$ such that $A + \varepsilon I_n$ is non-singular for all $\varepsilon \in (0, \delta)$. Hint: Express the statement with the help of quantifiers. For quantifiers, we have the following rules:

$$\neg(\exists x : P(x)) \Leftrightarrow (\forall x : \neg P(x)),$$

$$\neg(\forall x : P(x)) \Leftrightarrow (\exists x : \neg P(x)).$$

(a) if n is even, then $3n + 11$ is odd.

(b) if for every $\delta > 0$, there exists $\varepsilon \in (0, \delta)$, such that $A + \varepsilon I_n$ is singular, then A is non-singular

$\{ A \text{ is singular} \Leftrightarrow \text{non-sing}$
 $\exists \delta > 0, \forall \varepsilon \in (0, \delta): A + \varepsilon I_n \text{ is non sing.}$
 $\Leftrightarrow \forall \delta > 0, \exists \varepsilon \in (0, \delta): A + \varepsilon I_n \text{ is sing.}$

Contraposition
= contrap

$$A \rightarrow B \Leftrightarrow !B \rightarrow !A$$

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4. (8 points) Show the following three assertions with the specified proof strategy:

(a) (direct proof) For all integers a, b, c : If $a|b$ and $b|c$, then $a|c$. Here $x|y$ denotes that an integer x divides an integer y , i.e., there exists an integer q such that $x \cdot q = y$.

Assume $a|b$.

By definition, there exists an int k , that $b = a \cdot k \dots \textcircled{a}$

Assume $b|c$

By def., there exists an int. l , that $c = b \cdot l \dots \textcircled{b}$

Substitute $\textcircled{a} \rightarrow \textcircled{b}$

$$c = a \cdot k \cdot l^{\text{int}}$$

But $k \cdot l$ is an int, which means $a|c$

$\therefore$ The statement is proven directly.

(b) (proof by contrapositive) For all $x \in \mathbb{R}$: If $x^2 + 2x < 0$, then $x < 0$.

Assume $x \geq 0$

Contrapositive expression: If $x \geq 0$, then $x^2 + 2x \geq 0$.

$$\rightarrow x^2 + 2x \geq 0$$

$$x(x+2) \geq 0 \dots \textcircled{a}$$

However, $x \geq 0$ and $x+2 \geq 0 \rightarrow$ so the product $x(x+2) \geq 0 \dots \textcircled{b}$ $\swarrow \text{eq.}$

$\therefore$ The original statement is true.

(c) (proof by contrapositive) Let n be an integer. If n^2 is even, then n is even.

Assume n is odd.

By definition, $n = 2k+1$

$$n^2 = (2k+1)^2 = 4k^2 + 2k + 1$$

$$= 2(2k^2 + 1) + 1$$

even int. + 1 $\Rightarrow$ odd

$\therefore$ By contrapositive, if n^2 is even, then n is even.

(d) (proof by contradiction) For all real numbers x, y : If x is rational and y is irrational, then $x+y$ is irrational.

Assume $x \in \mathbb{Q}$ and $y \notin \mathbb{Q}$

Assume $x+y$ is rational

Let $x+y = k$, where $k \in \mathbb{Q}$

$$\rightarrow y = (x+y) - x$$

$$= k - x$$

$k-x \in \mathbb{Q}$

$k-x$ is rational, and this contradicts the assumption $y \notin \mathbb{Q}$.

$\therefore$ Our assumption is false, and $x+y$ must be irrational.